UNSW MARKETING ANALYTICS HACKATHON 2026

# Beyond a good story

What 1.45 million Kiva loan descriptions
can and cannot tell us about funding speed







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# Text carries forecast information, but no writing tip is proven

Summary across the explanatory models, robustness checks and forecast benchmark

EXPLANATORY MODELS

## 64 of 120

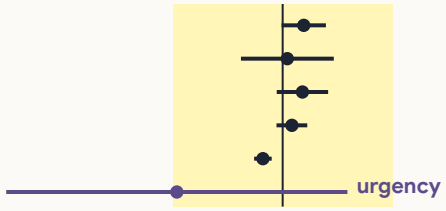


### Uncertainty first

Yellow squares: significant terms that lost significance once errors were clustered by 48 countries.

EQUIVALENCE TEST, EXPLORATORY

## ±5%

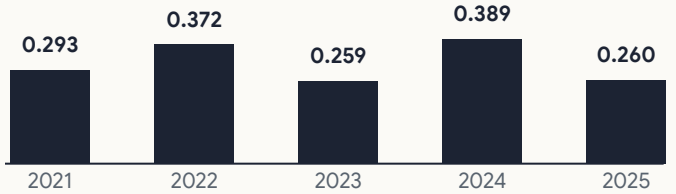



### Most framing links are small

Five framings stay inside the ±5% band per extra mention per 100 words; urgency does not.

FORECAST BENCHMARK

## 5 of 5 years




### Text adds forecast information

Text lowered linear forecast error every year, by 0.26–0.39 days; the median is about 7 hours.

**Next decision:** validate the text measures, then test one truthful prompt with partners.

Validate measures | Close data gaps | Pilot one prompt

# Agenda


01

## Problem

What are we measuring, and why does it matter?

- Editable fields 4
- The outcome 5
- A slower market 6

02

## Approach

How did we test the claims?

- Four analyses 7

03

## Evidence

What holds up, and how well?

- Country uncertainty 8
- Small or unknown 9
- Regional patterns 10
- Programme controls 11
- Forecast gain 12
- Limits of the gain 13

04

## Next steps

What should Kiva and partners do next?

- What is supported 14
- Three gates 15

# The description is one of the few listing fields still editable

Illustrative listing: what is fixed and what is editable when a loan is listed



**FIXED AT LISTING** 3

| Amount, median | Term, median | Borrowers, median |
|---|---|---|
| **US$325** | **12 months** | **1** |
| Sector, most common | Country, most common | Partner |
| **Agriculture** | **Philippines** | **Field partner** |

**EDITABLE DESCRIPTION** 1

She has a banana stall at the market to support her family. She wants to buy more stock quickly, before prices rise. She is thankful to every lender who helps her business grow. 4

Highlights mark framing terms counted per 100 words 2

Family | Business | Urgency | Gratitude

**Figure 1.** Illustrative composite listing, not a real borrower; values are dataset medians and most common categories.

### 1 The description is the editable part

Most fields are fixed. The **description can still be edited**, so if framing mattered, wording guidance would be a cheap lever.

### 2 Eight framing rates, every loan

**Eight framing rates per 100 words**, plus sentiment and text form, across 1,453,840 loans.

### 3 The test is for whom and when

Which framings are linked to **faster funding**, for which loans, and when?

### 4 Pronouns reflect who writes

Descriptions are mostly **written about the borrower, not by them**, so pronouns may reflect partner conventions.

# The outcome is platform funding time, not borrower waiting time

Descriptive audit of all records




**Loans disbursed to the borrower before listing**




**Who is in the data**

**In the data:** 1,453,840 completed loans, including 1,637 refunded

**Not in the data:** listings that expired or never funded 2

**Figure 2.** Journey of a typical pre-disbursed Kiva loan, not to scale; medians from the full-data audit.

[1] Typical field-partner flow; this step is not observed in the data.

[2] Separate medians: 24.2 days covers the 96.4% disbursed before listing, and the two do not add up along one loan.

**Scope:** 1,453,846 input records; 6 negative durations excluded; 1,637 refunded records included.

### 1 Partners wait, not borrowers

**96.4% of loans** reached the borrower a median 24.2 days before listing; platform time plausibly reflects replenishing partner capital.

### 2 Unfunded loans are missing

Only loans that **eventually funded** are in the data, so we cannot study whether a loan funds at all.

### 3 We measure platform speed

Our outcome runs from **listing to full funding**, median 2.85 days; it is not funding success or borrower welfare.

# Same-day funding fell from 46.0% to 30.3% and stayed low through 2024

Share of loans fully funded within 24 hours, by posting year; descriptive

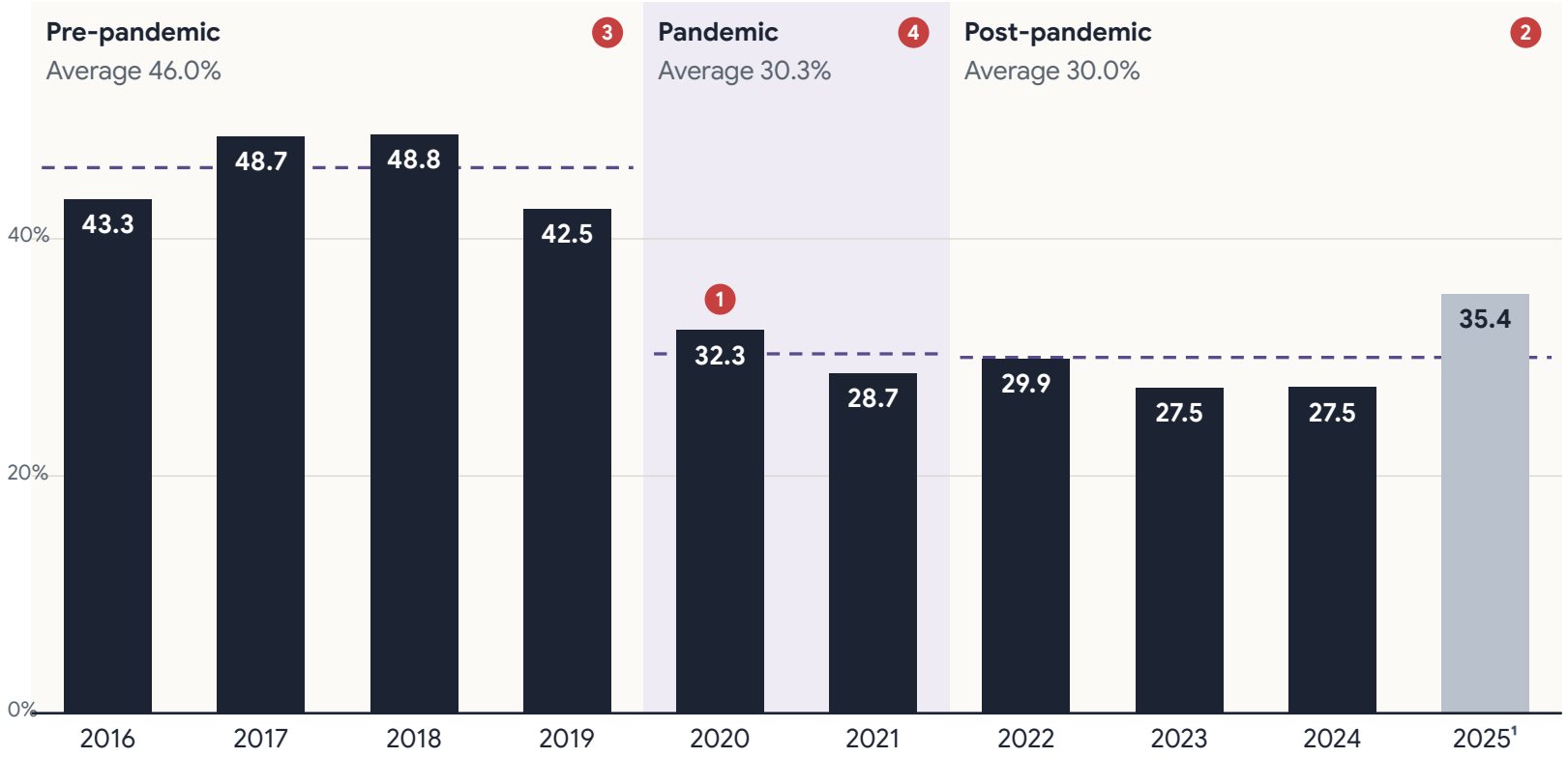



**Figure 3.** Share of completed loans funded within 24 hours, by posting year, with period averages.

[1] 2025 may be incomplete: loans still unfunded at extraction are missing.

### 1 The drop came in one year

Same-day funding fell from **42.5% in 2019 to 32.3% in 2020**.

### 2 The low level held four years

From 2021 to 2024 the share stayed **between 27.5% and 29.9%**; the 2025 rise may reflect an incomplete year.

### 3 Every model controls for period

The shift is platform-wide, so **narrative effects are compared within periods**, not across them.

### 4 Causes are untested here

Possible drivers, such as lender supply or listing volume, are **not measured** in this analysis.

# Four analyses, each answering a different question

One dataset; the equivalence test builds on the robustness checks, and the forecast benchmark asks a separate question

**Kiva loans, 2016–2025: 1,453,840 valid completed durations in 48 countries**

## Explanatory models → 8

**Which features are linked to funding speed, and how sure are we?**

Regressions of funding time and same-day funding; errors clustered by country.

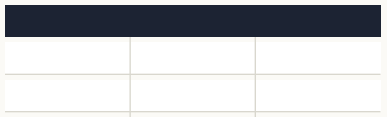


**64 of 120 significant terms lost significance**

## Robustness checks → 11

**Do the links survive richer controls?**

The same models with more text measures and a programme proxy; multiple-test adjusted.

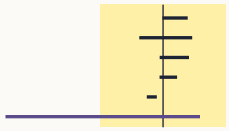


**Only small associations remain**

## Equivalence test → 9

**Which links are small enough to rule out a large effect?**

Is each 90% interval inside ±5% of funding time?



**Five framings small; urgency unresolved**

## Forecast benchmark → 12

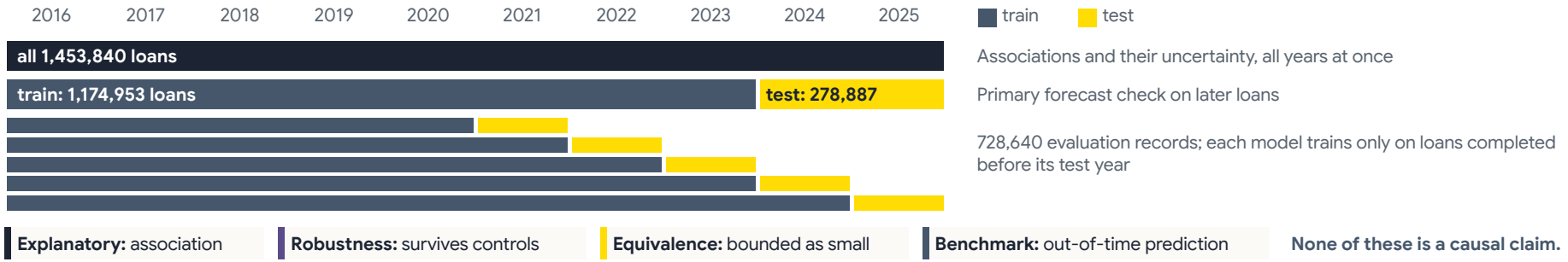
**Does text improve a forecast built on loan structure?**

Train on earlier years, then test each year with and without text.



**Error fell in 5 of 5 years, by 0.26–0.39 days**

### Years each analysis uses




**What each can claim**

- **Explanatory:** association
- **Robustness:** survives controls
- **Equivalence:** bounded as small
- **Benchmark:** out-of-time prediction

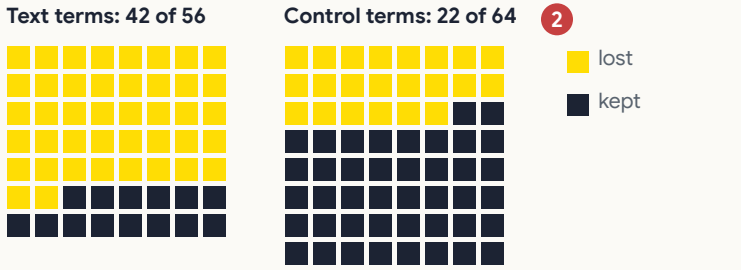
**None of these is a causal claim.**

# Large row counts did not remove country-level uncertainty

Explanatory models: HC3 versus 48-country clustered errors on 1,453,840 loans

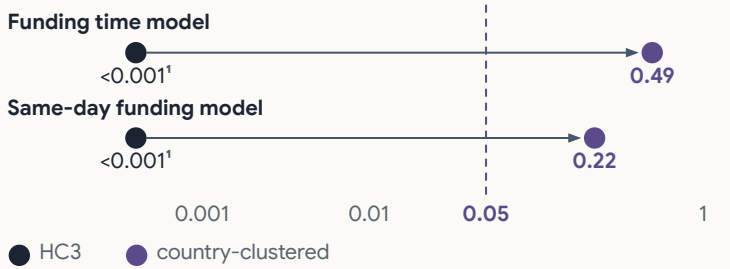
### (a) Which terms lost significance

64 of 120 terms significant under HC3 lost it under clustering; each square is one term.



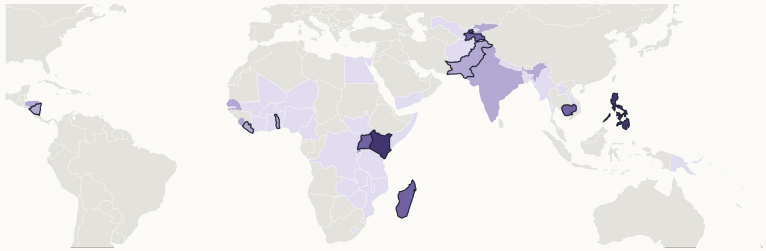

### (b) Urgency before and after

HC3 errors allow unequal variance but treat every loan as independent; clustering by country lets loans in a country share shocks.




### (c) Where the loans are

Ten of 48 countries hold 81% of loans (outlined on the map); table share cells use the map's colour scale.




| | Country | Loans | Share | Same-day |
|---|---|---|---|---|
| 1 | Philippines | 521,586 | 35.9% | 48% |
| 2 | Kenya | 266,562 | 18.3% | 29% |
| 3 | Uganda | 69,651 | 4.8% | 31% |
| 4 | Tajikistan | 68,865 | 4.7% | 18% |
| 5 | Cambodia | 58,074 | 4.0% | 49% |
| 6 | Madagascar | 55,670 | 3.8% | 67% |
| 7 | Pakistan | 42,767 | 2.9% | 15% |
| 8 | Nicaragua | 38,520 | 2.6% | 15% |
| 9 | Liberia | 28,847 | 2.0% | 28% |
| 10 | Togo | 27,513 | 1.9% | 38% |

**Figure 4.** Explanatory models, HC3 versus errors clustered by country (48 clusters); map and table: share of all loans and of loans funded within 24 hours, by country.

[1] Printed as 0.0000 in the saved run. Text terms include interactions of a text measure with a control.

### 1 Text terms are hit hardest

**42 of 56 text terms** lost significance, against 22 of 64 controls; 42 of the 56 terms that kept it are controls.

### 2 Few text terms survive

Only **sentiment and some family-framing interactions** with sector or region stay significant.

### 3 Urgency no longer passes

Its p-value rose from below 0.001 to **0.49** for funding time and 0.22 for same-day funding.

### 4 Few countries dominate

Ten countries hold **81% of loans**, and same-day funding ranges from 15% in Pakistan to 67% in Madagascar.

# Most framing associations are small; urgency remains unresolved

Equivalence test of the robustness-check coefficients; exploratory, bound set after the estimates

Shaded bands are three sizes of "too small to matter"; the primary ±0.05 is about ±5% of funding time, or 4.5–4.7 hours for a median loan. 3




**Figure 5.** Change in log(1 + days) per extra mention per 100 words, 90% intervals, expanded specification; groups use both specifications.

[1] Per +0.5 change in the sentiment score; no mention means a score of exactly zero.

**Note:** ±0.05 is the BH-adjusted primary bound; ±0.02 and ±0.10 are unadjusted sensitivity bounds.

### 1 Five framings are small

Five framings stay **inside ±5% of funding time** in both specifications, about **4.5–4.7 hours** for a median loan.

### 2 Urgency and first person are unknown

Their intervals still reach **6–14% faster or slower funding** per mention. Each is absent from **over 98% of records**, so the data say little about them.

### 3 The bound came after the data

It was chosen **after seeing the estimates**, so three bands are shown: family, agency and third person hold even at **±2%**.

### 4 The verdicts survive a stricter test

Swapping the t(47) reference for a stricter **t(4)** keeps **all five duration verdicts**.

# Regional family-framing patterns rest on two countries each

Explanatory models: within-region slopes with country-clustered and heuristic few-cluster intervals

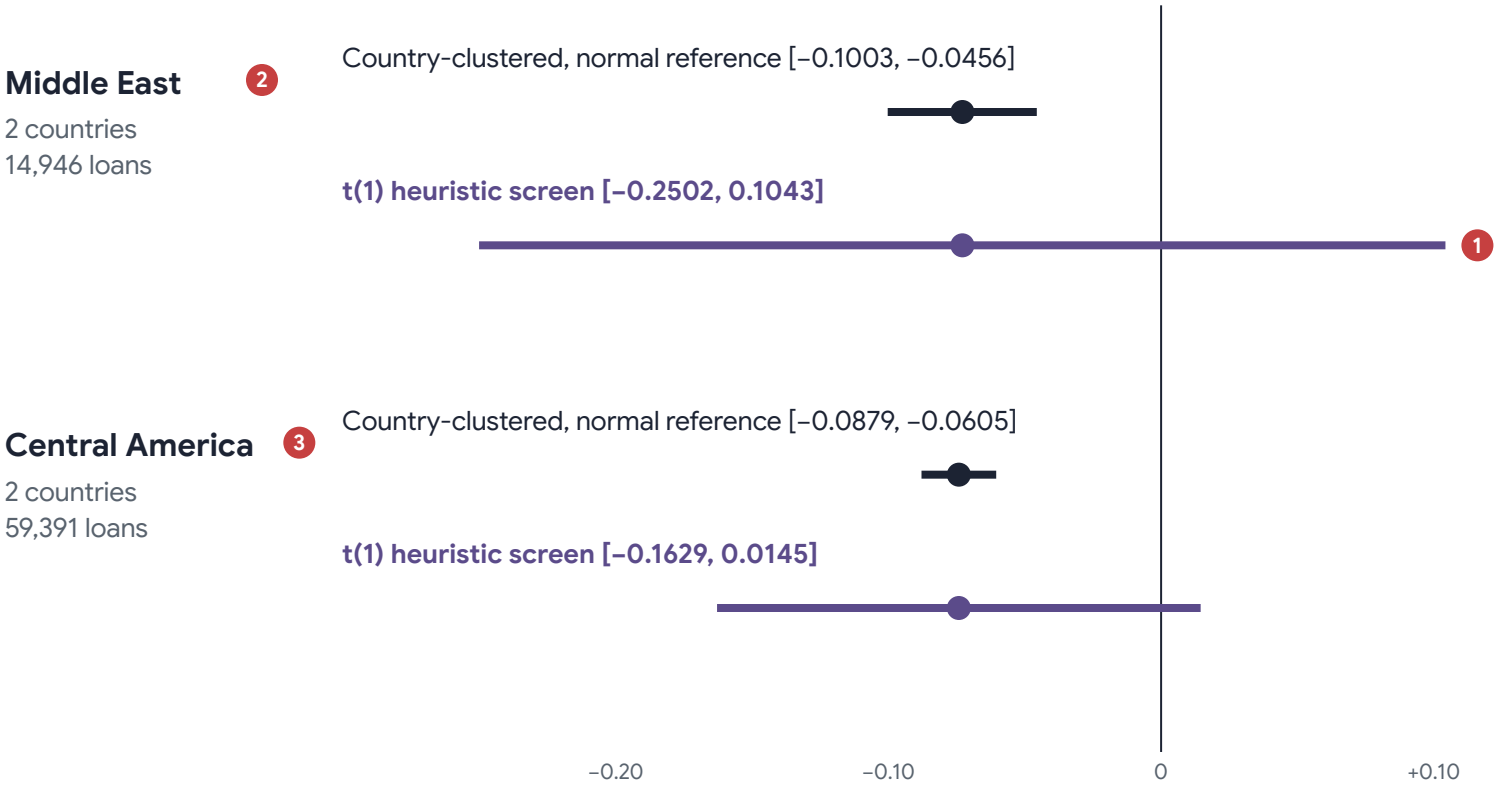



**Figure 6.** Within-region family-framing slopes, log(1 + days) per extra mention per 100 words.

### 1 Two countries cannot make a pattern

With two clusters, between-country uncertainty **cannot be estimated well**; the heuristic intervals cross zero.

### 2 No regional writing rules

These are descriptive patterns, **not grounds for country or regional writing rules**.

### 3 More countries would settle it

A regional test needs **more countries per region**, or partner-level data.

# Few narrative associations remain after programme controls

Robustness checks: country-clustered, BH-adjusted within specification

| Term and outcome | Expanded | Programme-adjusted |
|---|---|---|
| Third person, log(1 + days) | −0.0090, BH p 0.012 | −0.0081, BH p 0.061 |
| Business, 24-hour (pp) | not significant after BH | −0.258 pp, BH p 0.006 |

**Figure 7.** Per extra mention per 100 words; BH-adjusted within each specification.

**0.012 → 0.061** 3 1

third-person BH p, expanded to programme-adjusted

**−0.258 pp** 2

business framing, same-day rate, programme-adjusted

### 1 Only small associations remain

Third-person framing keeps its size but rises to **BH p 0.061**; business framing is **the one term below BH 0.05**.

### 2 These are still associations

Programme text is an **unvalidated proxy**; neither result shows what editing would do.

### 3 Pronouns may reflect the writer

Descriptions are usually written by partners, so the **third-person link may reflect who writes**, not a borrower's choice.

# Adding text lowered forecast error in 5 of 5 years

Forecast benchmark: linear model, structure only versus structure plus text, identical evaluation rows

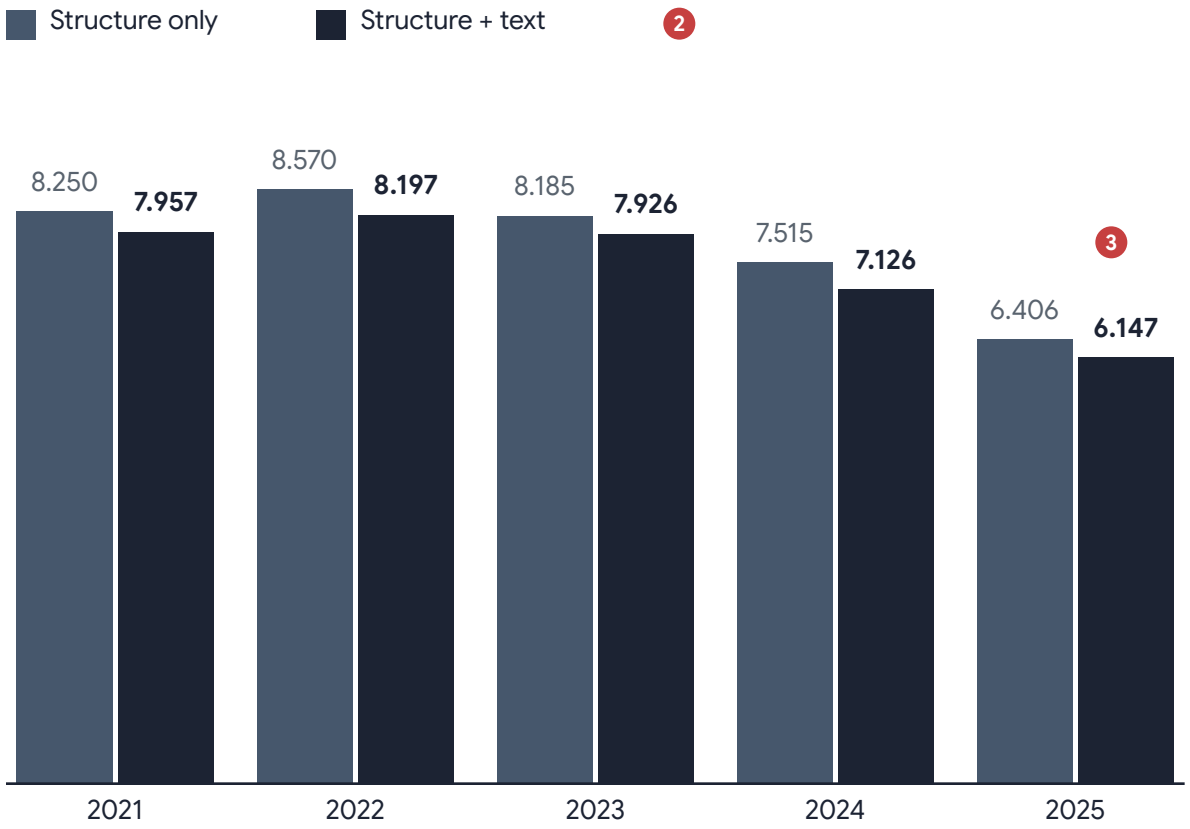



**Figure 8.** Linear-model MAE in days on identical evaluation rows; lower is better.

**0.29 days** 1

median annual error reduction

**5 of 5**

years with lower error

**728,640**

evaluation records

### 1 Text carries forecast information

Gains of **0.26–0.39 days** on 728,640 evaluation records.

### 2 Forecasting is not persuasion

Text may encode **purpose, place or programme templates**, not only persuasion.

### 3 No single country drives it

Removing any single country **leaves the gain positive** in every year (exploratory check).

# The gain is modest and depends on the model

Forecast benchmark: annual MAE gain with month-resampled 95% intervals

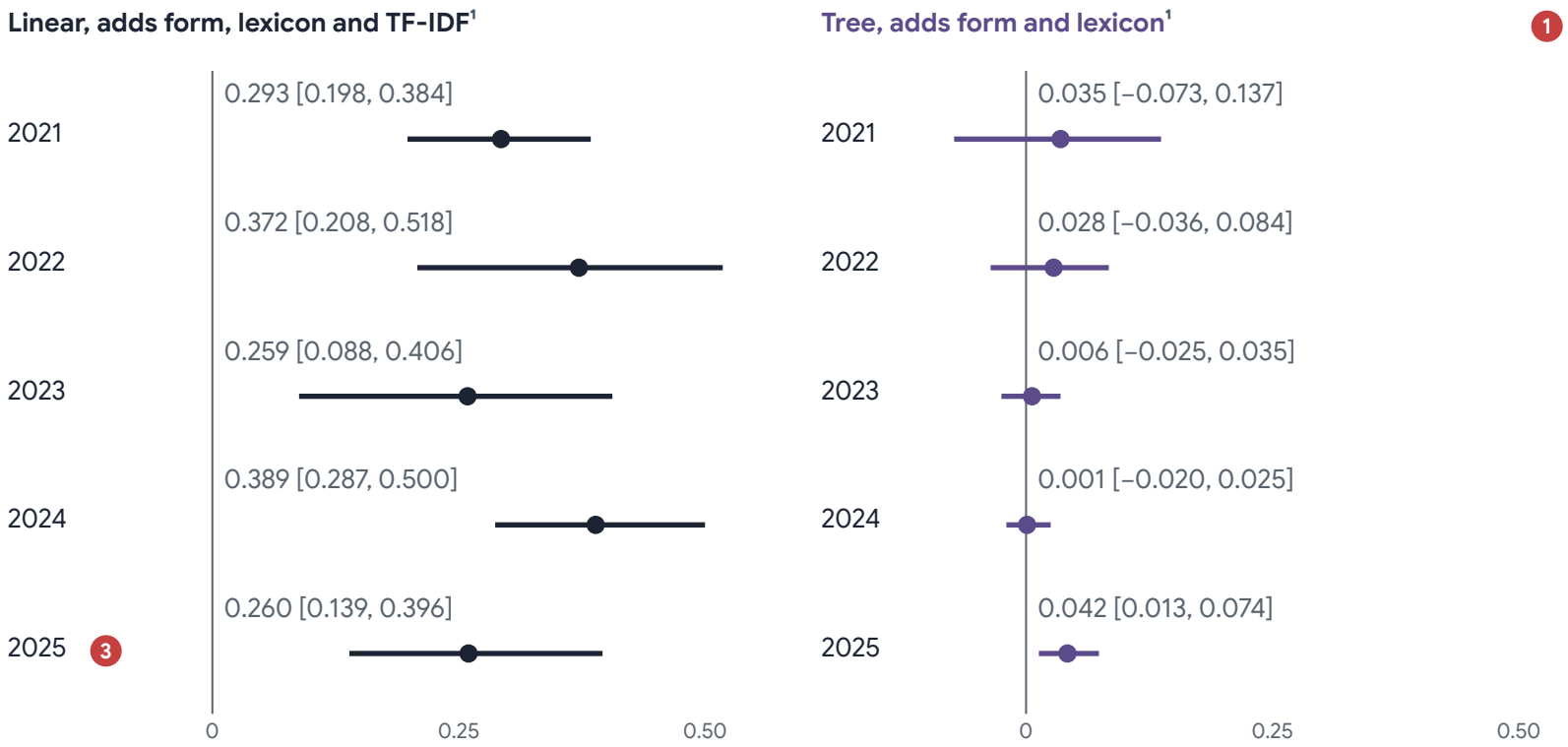



**Figure 9.** Annual MAE gain in days with month-resampled 95% intervals; positive is better.

2 [1] Country resampling, not shown: intervals include zero in 4 of 5 linear and 3 of 5 tree years.

### 1 The tree model gains little

The tree model, without TF-IDF, gains **at most 0.04 days**.

### 2 Country intervals are wide

Country resampling **includes zero in 4 of 5** linear years. Capping extreme forecasts, an exploratory check, leaves zero in 1 of 5.

### 3 Metrics move in different directions

In 2025 text improved error and calibration but **AUC fell** from 0.893 to 0.888.

# What the evidence supports, and what it does not

Synthesis across all layers

| Supported | Not established |
| --- | --- |
| Text adds forecast information in a linear model | That editing a description speeds funding |
| Most framing associations are small | Any country or regional writing rule |
| Platform funding time is not borrower waiting time for most loans | Effects of urgency or first-person framing |
| | Performance on loans that never funded |

# Three gates before any writing advice

Proposed next research for Kiva and participating partners; each gate must pass before the next

01 → 4

## Validate the measures

**Question**

**Do our word counts capture clarity, need and agency?**

**Do**

Build a small codebook; two people code the same narratives and resolve disagreements.

**Done when**

Definitions are frozen before later periods are checked.

02 → 5

## Close the data gaps

**Question**

**What did lenders see, and which listings never funded?**

**Do**

Get original texts, partner identifiers, exposure data and expired listings.

**Done when**

Unfunded listings and listing-time text are in the data.

03 → 14

## Pilot one prompt

**Question**

**Does editing a description change funding speed?**

**Do**

With partners, test one truthful clarity prompt against a control.

**Done when**

Threshold, power and guardrails are agreed in advance.

### Each gate unlocks a stronger claim

**Today**
Associations and forecast information

→

**After gate 1**
Measures that mean what we say

→

**After gate 2**
Results for every listing, not only funded ones

→

**After gate 3**
A tested effect of editing

**If a gate fails, stop there: report the uncertainty and give no writing advice.**

# Thank you

**Mike Nguyen and Sophia Nguyen, Cultural Blend**

Code and evidence: github.com/tuannm3812/unsw-ma-hackathon-2026



# The two-country screen in full

Explanatory models: within-region family-framing slopes in log(1 + days) per extra mention per 100 words

| Region | Countries | Loans | Slope | Clustered 95%, normal | t(1) interval | Heuristic p |
|---|---|---|---|---|---|---|
| Middle East | 2 | 14,946 | −0.0729 | [−0.1003, −0.0456] | [−0.2502, 0.1043] | 0.1203 |
| Central America | 2 | 59,391 | −0.0742 | [−0.0879, −0.0605] | [−0.1629, 0.0145] | 0.0597 |

**Figure A1.** Within-region family-framing slopes, log(1 + days) per extra mention per 100 words.

## Why two countries are not enough

A one-country group cannot estimate between-country uncertainty, and two countries estimate it poorly. The t(1) reference keeps the same clustered standard error and widens the interval.

## What the screen is

A heuristic sensitivity screen, not a calibrated correction. Passing it would not certify a result, so these patterns stay descriptive.

# As-of training audit

Forecast benchmark: training records must have completed before each cutoff

| Evaluation year | Pre-cutoff postings completing later | Eligible as-of training | Evaluation rows |
|---|---|---|---|
| 2021 | 7,528 | 717,672 | 163,166 |
| 2022 | 3,918 | 884,448 | 143,354 |
| 2023 | 3,546 | 1,028,174 | 143,233 |
| 2024 | 5,691[1] | 1,169,262 | 141,725 |
| 2025 | 4,261 | 1,312,417 | 137,162 |

**Figure A2.** Training availability and evaluation rows by year.

[1] A posting-only split would keep these 5,691 training records; evaluation windows still select eventual completions.

# Chronological holdout metrics

Explanatory models: chronological holdout, completed loans only

**5.2031 days**

duration MAE, HistGradientBoosting

**0.5420**

duration $R^2$

**0.8997**

24-hour ROC AUC

**0.8301**

24-hour average precision

**0.1156**

24-hour Brier score

**Scope:** 1,174,953 training and 278,887 holdout loans (2024–2025). Retrospective predictions on completed loans, not deployment estimates.

# Exploratory story themes

Exploratory NMF themes from a 20,000-loan sample; descriptive, not full-data evidence

## 1.5 days

mean funding time, fastest theme (sanitation)

## 13.5 days

mean funding time, slowest theme (group solar and farm plots)

### What the themes show

8 themes, including sanitation, clean water, livestock, smallholder farming and retail. They track loan purpose as much as writing style.

### How to read them

Means are unadjusted and come from a sample, so the gap is context, not a tested effect of story content.